

ProgressThesisJuly Part 3

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July 2026

1 Warnings

This document is written in a colloquial way.

2 Introduction

In this document I analyzed the congestion metric used to calibrate the demand data As I told you in a previous meeting a new congestion metric has been adopted.

3 Completed work

3.1 Congestion Metric

To explain this, I will include the docstring of the script that performs the analysis of the congestion metric.

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One-off tool: sweeps a range of demand levels, measures the congestion metric
for each, and automatically selects a representative demand value for each
congestion regime.
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```
Congestion metric
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The congestion metric is the average ratio of actual to free-flow trip duration:
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```
    congestion_metric = (1/N) * sum( T_i / T_i_free ) for i = 1..N agents
```

```
where T_i is the actual trip duration and T_i_free is the free-flow shortest-path
duration under an empty network. A value of 1.4 means trips take on average 40%
longer than free-flow. The metric is dimensionless and monotonically increasing
with congestion level.
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Free-flow travel times
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It should be noted, that the true free-flow travel time of an edge is not the same than
dividing the edge length by its maximum speed. Because the true free flow travel time
takes into account the acceleration, deceleration... Hence, the edge length / max speed, can
be seen as an estimate, but its not the true value. That estimate underestimates the true free
flow travel time value and hence it overestimates the congestion metric.
Because, even with an empty network, the congestion metric was taking values of 1.4~1.5, the true
free flow travel times were generated by running a simulation with almost no vehicles, and then the
ratio takes value of 1 for very low demands as it should.
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Congestion regimes and decision rule
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Two congestion regimes are defined (low, severe) via threshold lower bounds stored
in CONGESTION_REGIMES. These thresholds are qualitative and somewhat arbitrary
what matters is that they produce meaningfully distinct traffic states for the
experiments. For each regime, the representative demand is selected as the first
```

demand value where the metric exceeds the regime threshold and the immediately following demand value also stays above it. This two-point rule avoids picking a noisy spike that immediately drops back below the threshold.

The representative demands selected here are used throughout the thesis experiments to evaluate the MARL algorithm's performance under different congestion levels.

Demand grid

The demand sweep range (DEMANDS) must be set manually per network before running. The lower bound should start at or just below the onset of congestion, not at the free-flow baseline sweeping demands that produce identical free-flow results wastes simulation runs without adding information. The upper bound should be high enough to clearly reach heavy congestion, but not much beyond that: once the representative demand for the highest regime has been identified, additional points yield no useful information and only increase runtime. The step size trades resolution against runtime.

The script otherwise runs automatically: it sweeps demands, plots the congestion curve with regime bands and threshold lines, and marks the selected representative demands directly on the plot.

Run with: `python src/tools/sensitivity/plot_congestion_metric.py <config.yaml>`
"""

I have tried several random seeds to ensure that the obtained results are robust.

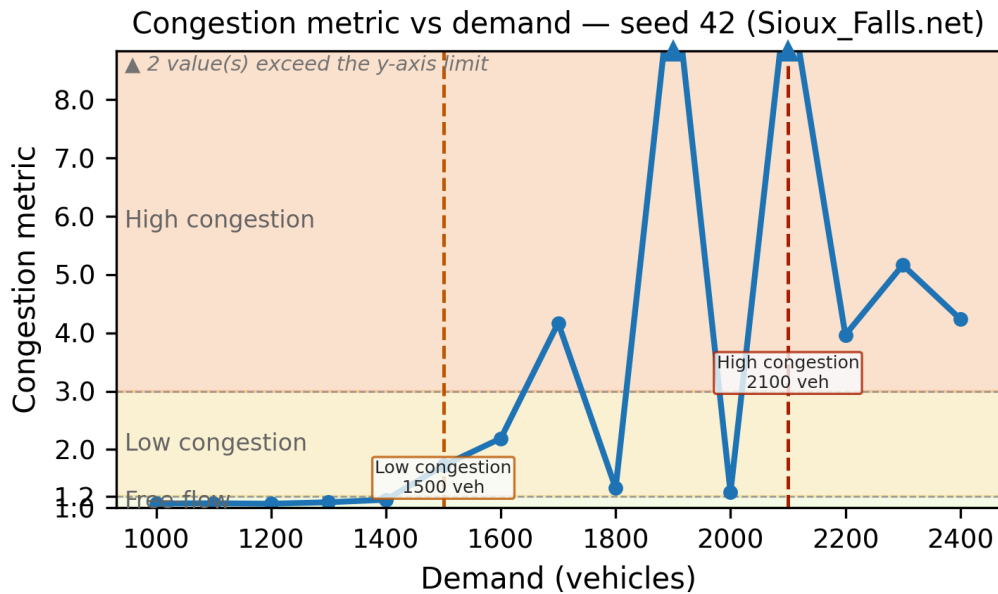


Figure 1: Congestion metric analysis. Seed 42

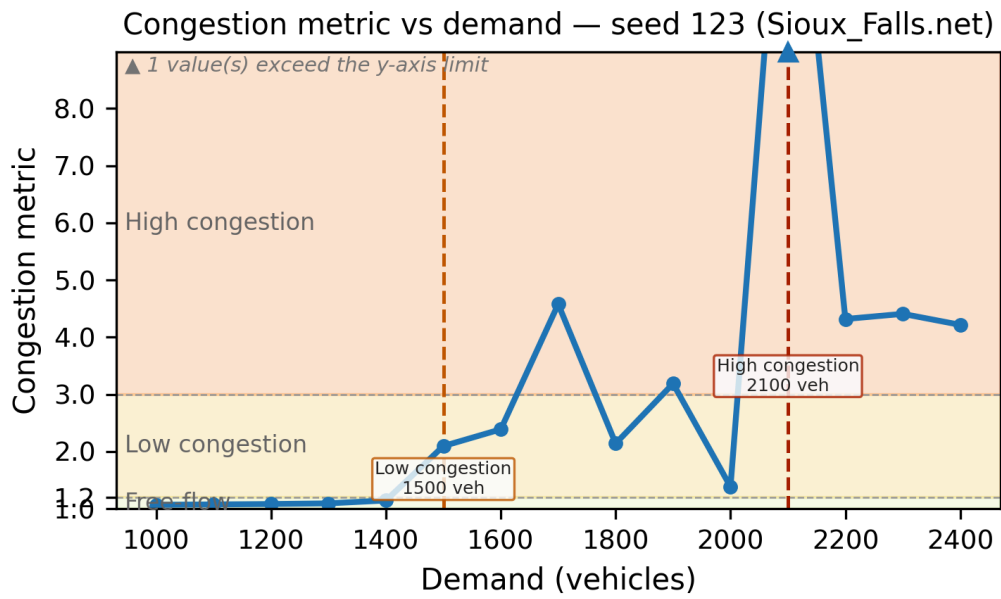


Figure 2: Congestion metric analysis. Seed 123

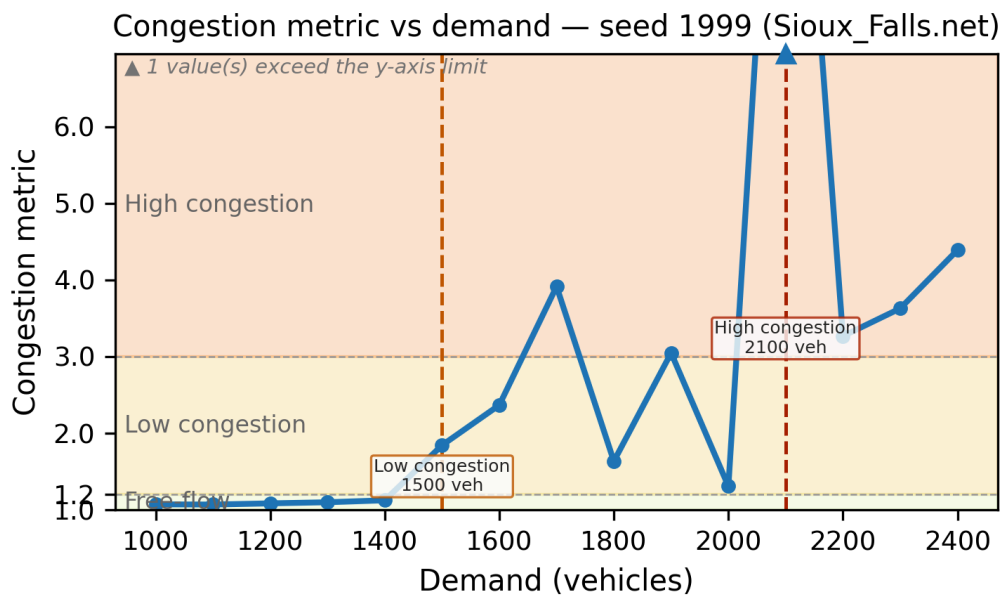


Figure 3: Congestion metric analysis. Seed 1999

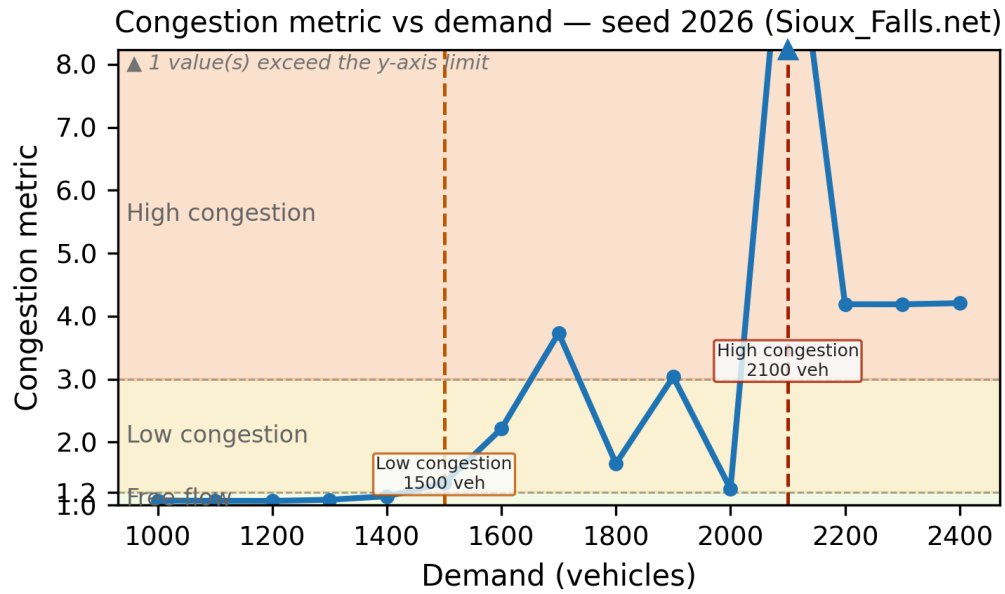


Figure 4: Congestion metric analysis. Seed 2026

Final decision Two congestion regimes are defined (low congestion and severe congestion). The representative demands values for each regime are chosen automatically with this script.

For the case of the Thesis with the Sioux Falls Network, the scenario of low congestion will be represented by a demand of 1500 veh. Whereas a scenario of high/severe congestion will be represented by a demand of 2100 vehicles.